

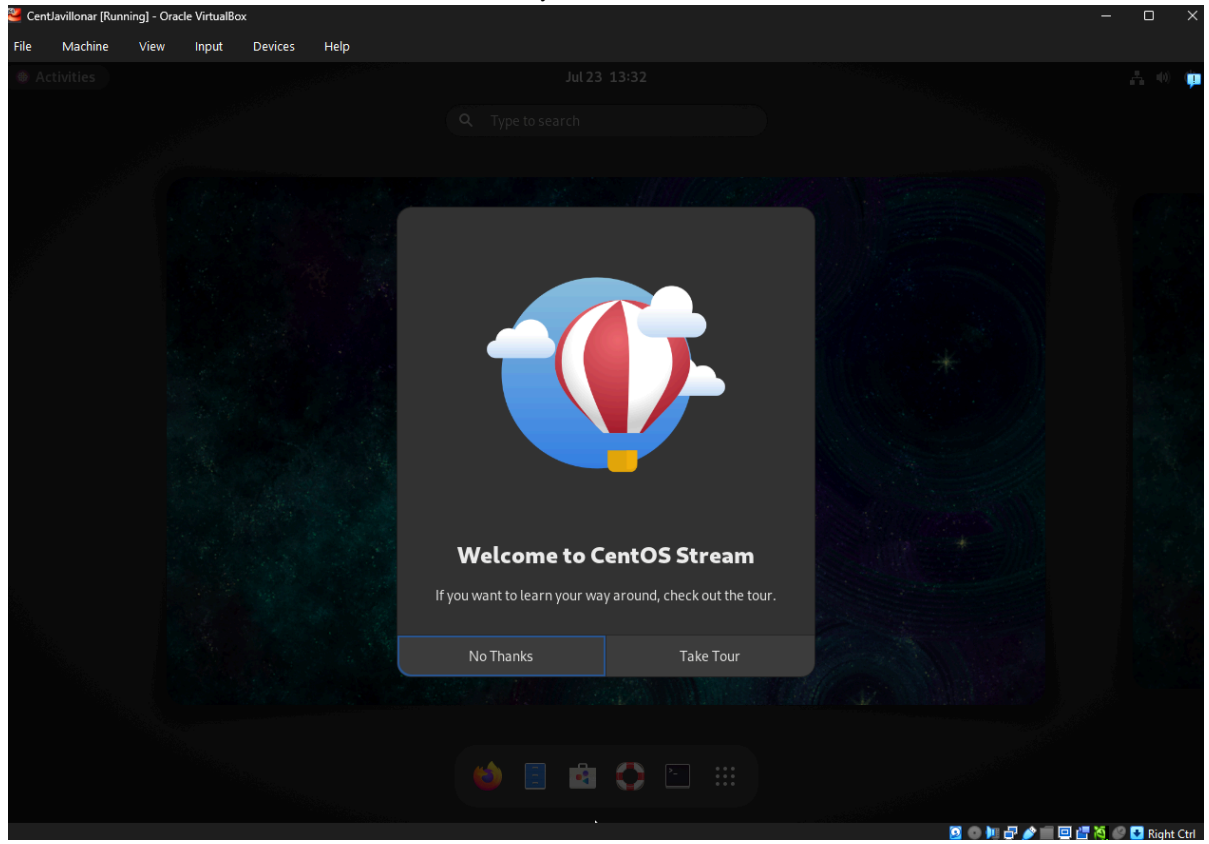
COURSE CODE/TITLE:	CPE 212- Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	7/23/2026	DATE SUBMITTED:	7/23/2026
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ACTIVITY NO. 3.1 Hands-on Activity - Install SSH server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here:
http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.



Task 2: Install the SSH server package openssh

1. Install the ssh server package openssh by using the dnf command:

```
$ dnf install openssh-server
```

2. Start the sshd daemon and set to start after reboot:

```
$ systemctl start sshd
```

```
$ systemctl enable sshd
```

3. Confirm that the sshd daemon is up and running:

```
$ systemctl status sshd
```

4. Open the SSH port 22 to allow incoming traffic:

```
$ firewall-cmd --zone=public --permanent --add-service=ssh
```

```
$ firewall-cmd --reload
```

5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes:

```
$ systemctl reload sshd
```

SCREENSHOTS:

```
[cjavillonar@localhost ~]$ sudo yum install openssh-server -y
[sudo] password for cjavillonar:
/var/cache/dnf/expired_repos.json is empty file
CentOS Stream 9 - AppStream          3.1 MB/s | 28 MB    00:09
CentOS Stream 9 - Extras packages    10 kB/s | 21 kB    00:02
Package openssh-server-9.9p1-10.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[cjavillonar@localhost ~]$ sudo dnf install openssh-server -y
Last metadata expiration check: 0:00:28 ago on Thu 23 Jul 2026 02:24:43 PM PST.
Package openssh-server-9.9p1-10.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[cjavillonar@localhost ~]$
```

```
[cjavillonar@localhost ~]$ sudo systemctl start sshd
[cjavillonar@localhost ~]$ sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: ena>
   Active: active (running) since Thu 2026-07-23 13:38:07 PST; 48min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 914 (sshd)
     Tasks: 1 (limit: 10500)
    Memory: 2.1M (peak: 2.3M)
       CPU: 27ms
    CGroup: /system.slice/ssh.service
           └─914 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 13:38:06 localhost.localdomain systemd[1]: Starting OpenSSH server daemo>
Jul 23 13:38:07 localhost.localdomain sshd[914]: Server listening on 0.0.0.0 po>
Jul 23 13:38:07 localhost.localdomain sshd[914]: Server listening on :: port 22.
Jul 23 13:38:07 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
[cjavillonar@localhost ~]$

[cjavillonar@localhost ~]$ sudo systemctl enable sshd
[cjavillonar@localhost ~]$ sudo systemctl is-enable sshd
Unknown command verb is-enable.
[cjavillonar@localhost ~]$ sudo firewall-cmd --zone=public --permanent --add-ser
vice=ssh
Warning: ALREADY_ENABLED: ssh
success
[cjavillonar@localhost ~]$ sudo firewall-cmd --reload
success
[cjavillonar@localhost ~]$ sudo firewall-cmd --list-services
cockpit dhcpv6-client ssh
[cjavillonar@localhost ~]$ cd /etc/ssh
[cjavillonar@localhost ssh]$ ls
moduli          sshd_config.d      ssh_host_ed25519_key.pub
ssh_config      ssh_host_ecdsa_key  ssh_host_rsa_key
ssh_config.d    ssh_host_ecdsa_key.pub  ssh_host_rsa_key.pub
sshd_config     ssh_host_ed25519_key
[cjavillonar@localhost ssh]$ pwd
/etc/ssh
[cjavillonar@localhost ssh]$
```

```

cjavillonar@localhost:/etc/ssh
cjavillonar@localhost ssh]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: ena>
   Active: active (running) since Thu 2026-07-23 13:38:07 PST; 57min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Process: 3657 ExecReload=/bin/kill -HUP $MAINPID (code=exited, status=0/SUC>
 Main PID: 914 (sshd)
    Tasks: 1 (limit: 10500)
  Memory: 2.1M (peak: 2.5M)
     CPU: 40ms
   CGroup: /system.slice/ssh.service
           └─914 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 13:38:06 localhost.localdomain systemd[1]: Starting OpenSSH server daemo>
Jul 23 13:38:07 localhost.localdomain sshd[914]: Server listening on 0.0.0.0 po>
Jul 23 13:38:07 localhost.localdomain sshd[914]: Server listening on :: port 22.
Jul 23 13:38:07 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
Jul 23 14:35:11 localhost.localdomain systemd[1]: Reloading OpenSSH server daem>
Jul 23 14:35:11 localhost.localdomain sshd[914]: Received SIGHUP; restarting.
Jul 23 14:35:11 localhost.localdomain sshd[914]: Server listening on 0.0.0.0 po>
Jul 23 14:35:11 localhost.localdomain sshd[914]: Server listening on :: port 22.
Jul 23 14:35:11 localhost.localdomain systemd[1]: Reloaded OpenSSH server daemo>

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Main PID: 914 (sshd)
  Tasks: 1 (limit: 10500)
Memory: 2.1M (peak: 2.5M)
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Jul 23 14:35:11 localhost.localdomain sshd[914]: Server listening on 0.0.0.0 po>
Jul 23 14:35:11 localhost.localdomain sshd[914]: Server listening on :: port 22.
Jul 23 14:35:11 localhost.localdomain systemd[1]: Reloaded OpenSSH server daemo>

cjavillonar@localhost ssh]$ sudo ss -tulpn | grep ssh
tcp  LISTEN 0      128          0.0.0.0:22      0.0.0.0:*      users:((("sshd",pid=914,fd=7))
tcp  LISTEN 0      128          [::]:22        [::]:*        users:((("sshd",pid=914,fd=8))

cjavillonar@localhost ssh]$ sudo netstat -tulpn | grep ssh
tcp        0      0 0.0.0.0:22        0.0.0.0:*        LISTEN      914/sshd: /usr/sbin
tcp6       0      0 :::22            :::*              LISTEN      914/sshd: /usr/sbin
cjavillonar@localhost ssh]$

```

Task 3: Copy the Public Key to CentOS

1. Make sure that ssh is installed on the local machine.
2. Using the command `ssh-copy-id`, connect your local machine to CentOS.
3. On CentOS, verify that you have the `authorized_keys`.

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.

SCREENSHOTS FOR TASK 3 and 4

```
C:\Users\user>ssh -V
OpenSSH_for_Windows_9.5p2, LibreSSL 3.8.2

C:\Users\user>ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\user\.ssh/id_ed25519):
Created directory 'C:\Users\user\.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\user\.ssh/id_ed25519
Your public key has been saved in C:\Users\user\.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:4rMtAp8ba/0ciJVwtTC9+xY0udDQ8yWrW3WKmVyHnBY user@DESKTOP-AB5D2JT
The key's randomart image is:
+--[ED25519 256]--+
|      o...      |
|    +o.o E .    |
|  . . .+ = B    |
|    o .o = O +  |
|  . S  = X +.   |
|  . + .. O o    |
|    =.. .. +    |
|  . oB.. .+     |
|  ..+*o..o.     |
+-----[SHA256]-----+
```

```
[cjavillonar@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:cc:de:73 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.23/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86366sec preferred_lft 86366sec
    inet6 fe80::a00:27ff:fecc:de73/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[cjavillonar@localhost ~]$
```

```
C:\Users\user>ip addr
'ip' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\user>ssh cjavillonar@192.168.1.23
The authenticity of host '192.168.1.23 (192.168.1.23)' can't be established.
ED25519 key fingerprint is SHA256:+a6Wj1Yfn4sLSn6KfvIR4Zu2gb6Qxqh4IDM14376pLM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.23' (ED25519) to the list of known hosts.
cjavillonar@192.168.1.23's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Jul 23 14:47:55 2026
[cjavillonar@localhost ~]$
```

```
C:\Users\user>dir C:\Users\user\.ssh
Volume in drive C has no label.
Volume Serial Number is 285A-0795

Directory of C:\Users\user\.ssh

07/23/2026  02:50 PM    <DIR>          .
07/23/2026  02:40 PM    <DIR>          ..
07/23/2026  02:41 PM             411 id_ed25519
07/23/2026  02:41 PM             103 id_ed25519.pub
07/23/2026  02:50 PM             837 known_hosts
07/23/2026  02:50 PM             95 known_hosts.old
               4 File(s)          1,446 bytes
               2 Dir(s)  51,124,256,768 bytes free

C:\Users\user>type %USERPROFILE%.ssh\id_ed25519.pub | ssh cjavillonar@192.168.1.2
3 "mkdir -p ~/.ssh && cat >> ~/.ssh/authorized_keys"
cjavillonar@192.168.1.23's password:
cjavillonar@localhost ~]$ ls ~/.ssh
authorized_keys
cjavillonar@localhost ~]$ act ~/.ssh/authorized_keys
bash: act: command not found...
Similar command is: 'cat'
cjavillonar@localhost ~]$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGiRxUiCUz1gbTq4VJCKQMFVEvVqBVAZfwsqJqgdMEsb
user@DESKTOP-AB5D2JT
cjavillonar@localhost ~]$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

- What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?
 - The best Linux distribution depends on its purpose. Important things to consider are how easy it is to use, its stability, the software it supports, and the available documentation. Debian is a good choice for learning Linux and personal use because it is free and reliable. Red Hat is a better choice for businesses because it provides professional support and enterprise features.
- What are the main differences between Debian and Red Hat Linux distributions?
 - Debian uses the APT package manager with .deb packages, while Red Hat uses DNF or YUM with .rpm packages. Debian is a free, community-supported Linux distribution, while Red Hat is a commercial distribution with paid support. Debian is commonly used for personal computers and servers, while Red Hat is widely used in companies because of its security, stability, and long-term support.

References:

Which OS is better? Red Hat (RHEL) vs. Debian. (2024, December 4). Ionos Digital Guide.

<https://www.ionos.com/digitalguide/server/know-how/red-hat-vs-debian/>

Meghwani, M., & Meghwani, M. (2025, November 7). Debian vs RHEL: Key Differences in Linux Distributions.

ServerAvatar | The first, Fully Hybrid Cloud Hosting Solution. <https://serveravatar.com/debian-vs-rhel/>

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

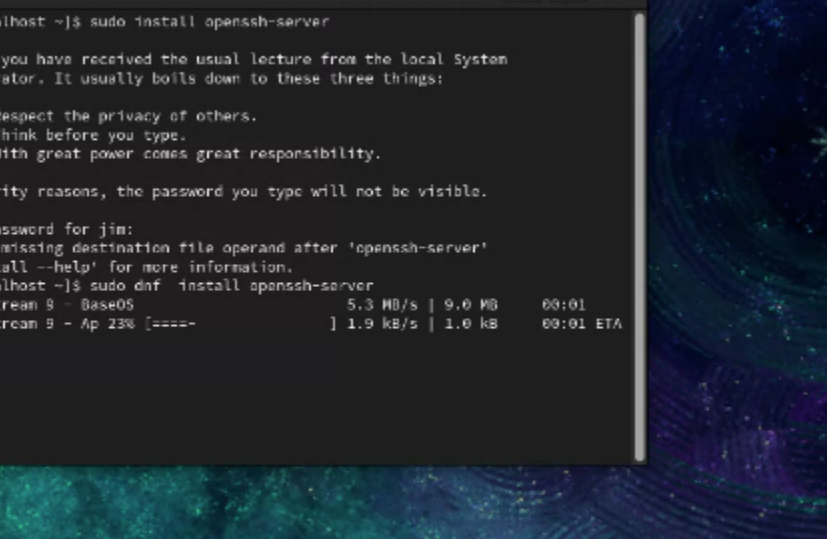
Extent of Use:

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```
jim@localhost:~ — sudo dnf install openssh-server
```

```
[jim@localhost ~]$ sudo install openssh-server
```

```
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:
```

```
#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.
```

```
For security reasons, the password you type will not be visible.
```

```
[sudo] password for jim:
```

```
install: missing destination file operand after 'openssh-server'
Try 'install --help' for more information.
```

```
[jim@localhost ~]$ sudo dnf install openssh-server
```

```
CentOS Stream 9 - BaseOS                    5.3 MB/s | 9.0 MB    00:01
CentOS Stream 9 - App Stream [-----] 1.9 kB/s | 1.0 kB    00:01 ETA
```

